Data augmentation project details

Self-organized pattern formation governs the arrangement and organization of biomolecules, cells and tissues that define the structure and function of biological organisms[1–3], across a wide range of spatial scales[4]. Microbial pattern formation, which constitutes the collective growth and migration of microorganisms across surfaces, often occurs on the centimeter scale[5–8]. Natural and engineered microbial systems can produce diverse spatial structures, including circular colonies[9–12], dendritic branching patterns[13–17], matrix-embedded biofilms[7,18], and multispecies range expansions[19–21]. Because these patterns occupy relatively large areas, experiments are often laborious and difficult to parallelize. This limited experimental throughput particularly becomes a bottleneck in building machine learning pipelines for forward or inverse problems—predicting patterns from input conditions[17] or inferring input conditions from observed patterns[17,22–24], respectively. Small training datasets can cause such models to overfit and generalize poorly across experimental conditions.

Traditional methods to handle data limitation often involve using heuristically selected transformations, for instance using the property of rotational invariance of circular plates to generate multiple samples from a single image. Recent studies have demonstrated augmentation using image generation models that can be guided using text[25] or simulation-conditioned images[17]. Although these methods can be useful in targeted applications, adapting such models across changes in bacterial species composition or environmental conditions would require significant methodology development.

An alternative approach that can address the gap is a replicate-conditioned, model-agnostic framework of data augmentation, wherein one biological replicate acts as the input to an image generation model, and another replicate across the same condition serves as the target output. Depending on the number of available replicates, the model could be trained using multiple input–output pairings and evaluated on held-out experimental conditions. Additionally functional tests could be assessed by determining whether the generated samples improve the performance and generalization of a downstream model. We hypothesize that, given sufficient replicates across diverse conditions, such a model could learn the variation observed within experimental datasets and provide more effective augmentation than conventional transformations.

Training the base data-augmentation model would require datasets spanning diverse microbial species and environmental conditions. Current and past research from our group can be a good starting point, including P. aeruginosa branching patterns across different species[11,15–17] (Lu et al, unpublished), environmental conditions[16] and combined with other species[6], or circular colony patterns in hard agar containing different species (Lok et al,

Villalobos et al unpublished). Other data from previously cited studies can be also extracted manually or in an automated manner. Diffusion models could serve as the initial baseline[26], with newer architectures, such as flow-based model[27,28], explored if further performance improvements are needed. Rather than immediately training on a large, combined dataset, an alternative strategy would be to first evaluate the framework by fine-tuning a pretrained model on smaller but diverse datasets.

Preliminary results

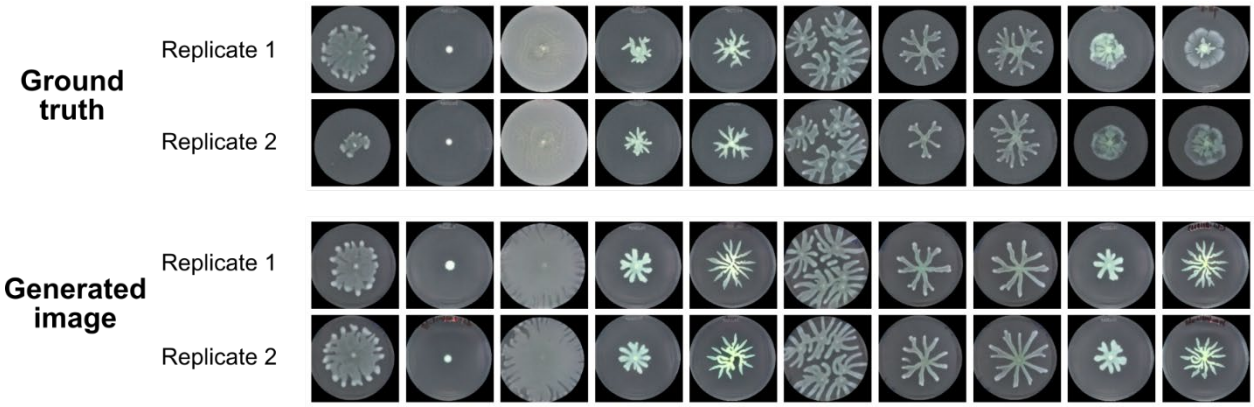


Figure 1: Preliminary performance evaluation of data augmentation model trained on 3 different datasets involving P. aeruginosa branching patterns

We trained a model on 3 different datasets- Sahu et al[17], Simsek et al[6] and Lu et al (unpublished). For increasing the data size to train the initial models, we first constructed all possible image pairs for available replicates, and then augmented using rotation (100 images per base image). This resulted in a total data size of 300,000 images. For inference, the model was tested on a test set that majority of samples of similar distribution as the training set. For testing out of distribution performance, we included some colony images that contained additional species, and the model did not estimate this well (last 2 columns in Figure 1 above). Judging by naked eye evaluation, the performance on other images is good. More metrics regarding the average image prediction performance need to be calculated for a better judgement.

References

1. Murray, J.D. (2007). Mathematical Biology: I. An Introduction (Springer Science & Business Media).

2. Kondo, S., and Miura, T. (2010). Reaction-Diffusion Model as a Framework for Understanding Biological Pattern Formation. Science *329*, 1616–1620.

3. Turing, A.M. (1952). The chemical basis of morphogenesis. Philos. Trans. R. Soc. Lond. B. Biol. Sci. *237*, 37–72. https://doi.org/10.1098/rstb.1952.0012.

4. Lu, J., Şimşek, E., Silver, A., and You, L. (2022). Advances and challenges in programming pattern formation using living cells. Curr. Opin. Chem. Biol. *68*, 102147.

5. Luo, N., Wang, S., Lu, J., Ouyang, X., and You, L. (2021). Collective colony growth is optimized by branching pattern formation in Pseudomonas aeruginosa. Mol. Syst. Biol. *17*, e10089. https://doi.org/https://doi.org/10.15252/msb.202010089.

6. Şimşek, E., Villalobos, C.A., Sahu, K., Zhou, Z., Luo, N., Lee, D., Ma, H.R., Anderson, D.J., Lee, C.T., and You, L. (2025). Spatial proximity dictates bacterial competition and expansion in microbial communities. Nat. Commun. *16*, 10885. https://doi.org/10.1038/s41467-025-65892-9.

7. Chou, K.-T., Lee, D.D., Chiou, J., Galera-Laporta, L., Ly, S., Garcia-Ojalvo, J., and Süel, G.M. (2022). A segmentation clock patterns cellular differentiation in a bacterial biofilm. Cell *185*, 145-157.e13. https://doi.org/10.1016/j.cell.2021.12.001.

8. Kim, H., Skinner, D.J., Glass, D.S., Hamby, A.E., Stuart, B.A.R., Dunkel, J., and Riedel-Kruse, I.H. (2022). 4-bit adhesion logic enables universal multicellular interface patterning. Nature *608*, 324–329. https://doi.org/10.1038/s41586-022-04944-2.

9. Kayser, J., Schreck, C.F., Gralka, M., Fusco, D., and Hallatschek, O. (2019). Collective motion conceals fitness differences in crowded cellular populations. Nat. Ecol. Evol. *3*, 125–134. https://doi.org/10.1038/s41559-018-0734-9.

10. Kannan, H., Sun, H., Warren, M., Çağlar, T., Yao, P., Taylor, B.R., Sahu, K., Ge, D., Mori, M., Kleinfeld, D., et al. (2025). Spatiotemporal development of expanding bacterial colonies driven by emergent mechanical constraints and nutrient gradients. Nat. Commun. *16*, 4878. https://doi.org/10.1038/s41467-025-60004-z.

11. Lu, J., Luo, N., Liu, S., Sahu, K., Maddamsetti, R., Baig, Y., and You, L. (2024). Discovery of interpretable patterning rules by integrating mechanistic modeling and deep learning. Preprint at bioRxiv, https://doi.org/10.1101/2024.09.02.610872 https://doi.org/10.1101/2024.09.02.610872.

12. Park, J.H., Holló, G., and Schaerli, Y. (2024). From resonance to chaos by modulating spatiotemporal patterns through a synthetic optogenetic oscillator. Nat. Commun. *15*, 7284. https://doi.org/10.1038/s41467-024-51626-w.

13. Ben-Jacob, E., Cohen, I., and Gutnick, D.L. (1998). COOPERATIVE ORGANIZATION OF BACTERIAL COLONIES: From Genotype to Morphotype. Annu. Rev. Microbiol. *52*, 779–806. https://doi.org/10.1146/annurev.micro.52.1.779.

14. Xiong, L., Cao, Y., Cooper, R., Rappel, W.-J., Hasty, J., and Tsimring, L. (2020). Flower-like patterns in multi-species bacterial colonies. eLife *9*, e48885. https://doi.org/10.7554/eLife.48885.

15. Luo, N., Lu, J., Simsek, E., Silver, A., Yao, Y., Ouyang, X., West, S.A., and You, L. (2024). The collapse of cooperation during range expansion of Pseudomonas aeruginosa. Nat Microbiol. https://doi.org/10.1038/s41564-024-01627-8.

16. Luo, N., Wang, S., Lu, J., Ouyang, X., and You, L. (2021). Collective colony growth is optimized by branching pattern formation in Pseudomonas aeruginosa. Mol. Syst. Biol. *17*, e10089. https://doi.org/https://doi.org/10.15252/msb.202010089.

17. Sahu, K., Davis, H.M., Lu, J., Villalobos, C.A., Heyman, A., Şimşek, E., and You, L. (2026). Simulation-conditioned generative modeling for biologically realistic pattern prediction. Preprint at bioRxiv, https://doi.org/10.64898/2025.12.13.694038 https://doi.org/10.64898/2025.12.13.694038.

18. Nadell, C.D., Drescher, K., and Foster, K.R. (2016). Spatial structure, cooperation and competition in biofilms. Nat. Rev. Microbiol. *14*, 589–600. https://doi.org/10.1038/nrmicro.2016.84.

19. Goldschmidt, F., Caduff, L., and Johnson, D.R. (2021). Causes and consequences of pattern diversification in a spatially self-organizing microbial community. ISME J. *15*, 2415–2426. https://doi.org/10.1038/s41396-021-00942-w.

20. Curatolo, A.I., Zhou, N., Zhao, Y., Liu, C., Daerr, A., Tailleur, J., and Huang, J. (2020). Cooperative pattern formation in multi-component bacterial systems through reciprocal motility regulation. Nat. Phys. *16*, 1152–1157. https://doi.org/10.1038/s41567-020-0964-z.

21. Liu, C., Fu, X., Liu, L., Ren, X., Chau, C.K.L., Li, S., Xiang, L., Zeng, H., Chen, G., Tang, L.-H., et al. (2011). Sequential Establishment of Stripe Patterns in an Expanding Cell Population. Science *334*, 238–241. https://doi.org/10.1126/science.1209042.

22. Doshi, A., Shaw, M., Tonea, R., Moon, S., Minyety, R., Doshi, A., Laine, A., Guo, J., and Danino, T. (2023). Engineered bacterial swarm patterns as spatial records of environmental inputs. Nat Chem Biol *19*, 878–886. https://doi.org/10.1038/s41589-023-01325-2.

23. Shaw, M., Garg, S., Kirby, J., Schmidgall, S., Liguori, F., Doshi, A., Tu, T., and Danino, T. (2026). Engineered E. coli swarming for binary and analog input recording. Mol. Syst. Biol. https://doi.org/10.1038/s44320-026-00232-7.

24. Lu, J., Tsoi, R., Luo, N., Ha, Y., Wang, S., Kwak, M., Baig, Y., Moiseyev, N., Tian, S., and Zhang, A. (2022). Distributed information encoding and decoding using self-organized spatial patterns. Patterns *3*.

25. Yang, Y., Li, W., Liu, R., Wu, C., Ren, J., Shi, Y., and Ge, S. (2025). HindwingLib: A library of leaf beetle hindwings generated by Stable Diffusion and ControlNet. Sci. Data *12*, 680. https://doi.org/10.1038/s41597-025-05010-y.

26. Zhang, L., Rao, A., and Agrawala, M. (2023). Adding Conditional Control to Text-to-Image Diffusion Models. In Proceedings of the IEEE/CVF international conference on computer vision, pp. 3836–3847. https://doi.org/10.48550/arXiv.2302.05543.

27. Lipman, Y., Havasi, M., Holderrieth, P., Shaul, N., Le, M., Karrer, B., Chen, R.T.Q., Lopez-Paz, D., Ben-Hamu, H., and Gat, I. (2024). Flow Matching Guide and Code. Preprint at arXiv, https://doi.org/10.48550/arXiv.2412.06264 https://doi.org/10.48550/arXiv.2412.06264.

28. Zhang, Y., Su, Y., Wang, C., Li, T., Wefers, Z., Nirschl, J., Burgess, J., Ding, D., Lozano, A., Lundberg, E., et al. (2025). CellFlux: Simulating Cellular Morphology Changes via Flow Matching. Preprint at arXiv, https://doi.org/10.48550/arXiv.2502.09775 https://doi.org/10.48550/arXiv.2502.09775.